MILITARY SPECIFICATION

CABLE ASSEMBLIES, RADIO FREQUENCY (AIRCRAFT AUDIO), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for complete cable assemblies, with molded or attached plugs including any required built-in electronics, used in aircraft intracommunications and intercommunications systems. It covers cord sets for personal gear such as microphones, headsets, earphones, handsets and similar equipment, and cable assemblies for contact with ground crew personnel.

1.2 Part numbers. Part numbers for cable assemblies furnished under this specification shall be as follows: M22442/X-Y with "X" being the specification sheet number and "Y" being any type cable assembly on a specific specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-J-641/22	Jack, Telephone U-385/U
MIL-P-642/14	Plug, Telephone (Type U-384/U)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military (Continued)

MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-26482	Connectors, Electrical (Circular, Miniature, Quick Disconnect, Environmental Resisting), Receptacles and Plugs, General Specification for
MIL-C-39029	Contacts, Electrical Connector, General Specification for
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

(See Supplement 1 for list of associated specifications.)

STANDARDS

Federal

FED-STD-228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
DOD-STD-2000	General Purpose Soldering Requirements for Electrical and Electronic Assemblies
MIL-STD-1353	Electrical Connectors, Plug-In Sockets and Associated Hardware, Selection and Use of
MIL-STD-45662	Calibration System Requirements

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SD-6	Provisions Governing Qualification

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D1149-81 Accelerated Ozone Cracking of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORY, INC.

UL-1581 Reference Standard for Electrical Wire, Cables and Flexible Cords

(Application for copies should be addressed to the Underwriters Laboratory, Inc., Publications Stock, 333 Pfingsten Road, North Brook, IL 60062.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Classification of requirements. The requirements for the cable assemblies are classified herein as follows:

Requirement	Paragraph
Qualification	3.3
Design and Construction	3.4
Performance	3.5
Environmental	3.6

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* 3.3 Qualification. Cable Assemblies furnished under this specification shall be products which are tested and qualified for listing on the applicable Qualified Products List (QPL) (see 4.4 and 6.3). In the event that no vendors for a cable assembly are listed on the current QPL, then the procuring activity shall require First Article testing equivalent to Qualification testing, prior to delivery of any production units. The qualifying activity shall have cognizance over any first article tests performed under this specification.

3.3.1 Qualification by similarity. Cable assemblies furnished under this specification may be qualified by similarity as specified in MIL-C-22442/6.

3.4 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400, Class I, for design and construction unless otherwise specified herein or in the specification sheet.

3.4.1 Connectors. The selection of connectors shall be in accordance with MIL-STD-1353 or as specified in the specification sheet.

3.4.1.1 Contacts. Contacts used in MIL-C-26482 type circular connectors shall be in accordance with MIL-C-39029. Other contacts shall be as specified in the specification sheet.

3.4.2 Connections. Connections shall be as specified in the specification sheet.

3.4.3 Terminations. Terminations shall be in accordance with MIL-STD-454 Requirement 19.

3.4.4 Telephone jacks. Telephone jacks shall conform to MIL-J-641/22 unless otherwise specified in the specification sheet.

3.4.5 Telephone plugs. Telephone plugs shall conform to MIL-P-642/14 unless otherwise specified in the specification sheet.

3.4.6 Cable (cord). Cable (cord) shall be in accordance with MIL-STD-454 Requirement 71 or as specified in the specification sheet. Cable sizing, configuration, and number of conductors shall be as specified in the specification sheet.

3.4.7 Extended length. Extended length of the cable assembly shall be as specified in the specification sheet.

3.4.8 Weight. The maximum weight of the cable assembly shall be as specified in the specification sheet.

3.4.9 Switch assemblies. Each switch assembly housing shall be built in accordance with the specification sheet and may contain a microphone amplifier, microphone keying switch, earphone impedance matching transformer, and terminal block, or any combination thereof, as specified in the specification sheet.

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3.4.10 Switches. Switches shall be as specified in the specification sheet.

3.4.11 Deleted.

3.4.12 Terminal blocks. Terminal blocks shall be as specified in the specification sheet.

3.4.13 Transformers. Deleted.

3.4.14 Aluminum alloys. Where aluminum alloy parts are employed they shall be cleaned by immersion in a suitable solution until etched, rinsed thoroughly in clean cold water, immersed in nitric acid until the remaining caustic is neutralized, and thoroughly rinsed in cold, then hot water. Anodic treatment is not prohibited, but masking shall be employed where necessary to retain the quality of the original metal in areas where good electrical contact is necessary.

3.4.15 Other non-ferrous metals and alloys. Copper, Brass, Bronze, Beryllium-Copper and other similar materials shall be plated or otherwise suitably finished or enclosed to provide protection against corrosion and deterioration. The kind of plating or finish applied shall depend upon the characteristics of the individual part and its use in the cable assembly. The plating or finish used for each individual part shall be such as to provide good appearance, corrosion and deterioration resistance, and long life for that part. Various problems such as soldering, fitting, assembly, reaction with adjacent parts and the like, must be considered in the selection of the plating or finish.

3.4.16 Dissimilar metals. Dissimilar metal combinations as specified in MIL-STD-454, requirement 16 shall not be used.

3.4.17 Interchangeability. Cable assemblies shall meet the interchangeability requirements of MIL-E-5400.

3.4.18 Deleted.

* 3.4.19 Marking. Cable assemblies shall be marked on plastic bands, or heat shrinkable tubing, and shall fit snugly around the cable circumference. If required, more than one identification band may be used. Heat shrinkable tubing when used shall not be used as support or as a part of the cable. The cable assemblies shall be marked with the following information.

Information Required	Example
Contract Number	****
Military P/N (Type Designator)	M22442/XX-X (CX-XXXX)
Contractor and Government Entities (Cage)	*****
NSN	NSN 5995-***

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3.4.20 Color. Color of the cable shall be black.

3.4.21 Clothing Clip. The clothing clip shall be as specified in the specification sheet.

3.4.22 Insulating and impregnating compounds. All such compounds including varnishes, waxes and the like, shall be suitable in every respect for each particular application. Under all service conditions, including test conditions outlined in Section 4 of this specification, the compound shall adequately preserve the dielectric strength of the insulation to which it is applied. A compound shall not, either in the state of its original application or as a result of aging under severe service conditions, show any injurious effect upon the insulation it is designed to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.4.23 Cementing compounds. Any cementing compound used in the construction of the cable assembly shall be of such a character and quality that it will withstand all service conditions including test conditions outlined in Section 4 of this specification without evidence of loosening or otherwise affecting the performance requirements of the cable assembly.

3.4.24 Soldering. Soldering shall be in accordance with the general purpose soldering requirements of DOD-STD-2000.

3.5 Performance. The cable assembly shall meet the following performance requirements when applicable.

* 3.5.1 Deleted (see MIL-C-22442/1).

* 3.5.1.1 Deleted (see MIL-C-22442/1).

* 3.5.1.2 Deleted (see MIL-C-22442/1).

* 3.5.1.3 Deleted (see MIL-C-22442/1).

* 3.5.1.4 Deleted (see MIL-C-22442/1).

* 3.5.2 Deleted (see MIL-C-22442/1).

* 3.5.2.1 Deleted (see MIL-C-22442/1).

* 3.5.2.2 Deleted (see MIL-C-22442/1).

* 3.5.2.3 Deleted (see MIL-C-22442/1).

* 3.5.2.4 Deleted (see MIL-C-22442/1).

* 3.5 2.5 Deleted (see MIL-C-22442/1).

* 3.5.2.6 Deleted (see MIL-C-22442/1).

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3.5.3 Continuity. Each conductor shall be continuous in the assembly through the terminations of the conductors, where the design calls for a continuous line. This shall include continuity across wire terminal junctions. The maximum dc resistance of circuit paths shall be as specified in the specification sheet.

3.5.4 Deleted.

* 3.5.5 Dielectric withstanding voltage. The cable assemblies shall be capable of withstanding a dielectric withstanding voltage as specified in the applicable cordage specification for a minimum of 30 seconds unless otherwise specified.

3.5.6 Gauging. For connector terminated cable assemblies, contacts and interfacial dimensions shall not be changed by the assembly procedure to prevent intermating of connectors.

* 3.5.7 Pull and twist. The terminal components (metallic terminals, taper pins, connectors, terminal blocks, molded bend reliefs) shall not part from the conductor at the specified loads. After twisting and pulling (see 4.6.9), there shall be no more than 1/32 inch slippage of the cable jacket from the connector, nor separation of the bond on molded terminations, molded bend reliefs, and molded strain reliefs.

3.5.8 Deleted.

3.5.9 Deleted.

3.5.10 Bend relief flexibility (see 6.5). For flexible molded cable assemblies with bend reliefs, there shall be no electrical discontinuity nor bond separation or cracking between the insulation or bend relief and the cable jacket after being tested in accordance with 4.6.12.

3.5.11 Low temperature flexibility. For cable assemblies with molded or heat-shrunken components, the insulation of each conductor within a portion of the cable jacket containing a molded or potted splice, crotch, or other molded or potted section or heat-shrunken transition (see 6.5), shall be tested for low temperature flexibility. The condition of the test temperature shall be the lower limit of the cable operating temperature range. If not specified, the temperature shall be -40°C. After test, there shall be no visible cracks or other damage in the molded section or adjacent portion of the cable jacket, and the assembly shall meet the specified performance.

3.5.12 Deleted.

3.5.13 Durability. The cable connectors shall exhibit a contact resistance and maximum mating force or torque as specified in the applicable drawings or specifications. After the cable assembly and connector have been subjected to the durability test, connectors and cable assemblies shall not show signs of degradation of engagement or separation forces, wear of contact plating or coupling mechanisms when tested as specified in the applicable drawings or specifications.

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3.5.14 Ozone resistance. Cable assemblies shall be tested for ozone resistance. At the completion of the exposure period, there shall be no evidence of cracking in the bend relief, molded junction or cable jacket when examined under 7X magnification.

3.5.15 Jacket elongation. The jacket elongation requirement shall be verified on each lot of cable used on the contract within 12 months of the date of shipment to the government. The test requirement shall be governed by the applicable cable jacketing specification for the section under test. The cable jacketing specification shall be as specified in the applicable MIL-C-22442 specification sheet. If no elongation requirements are given, the requirement shall be 175% minimum.

3.5.16 Durometer hardness. The cable assemblies shall be capable of meeting the durometer hardness of 60 ± 10 shore "A". This shall apply to all molded terminations and other molded sections.

3.5.17 Material reversion (Polyurethane and Polyacrylic Compounds Only). Molding, potting, insulating, or jacketing compounds shall not crack, run, drip, or deform, and the hardness after conditioning shall be no more than 10 points below the durometer hardness Shore "A" measured by 3.5.16; for polyurethane compounds cured with non-carcinogenic agents, the change shall be no more than 15 points.

3.6 Environmental.

3.6.1 Temperature - altitude. Cable assemblies with electronic components shall meet the performance requirements of the applicable specification sheet during and after the temperature-altitude test of 4.7. The cable assembly shall be operational when specified by MIL-T-5422. The steps performed shall be in the sequence specified in 4.7.

3.6.2 Salt fog. After completion of the salt fog test of 4.7, the cable assemblies shall show no evidence of corrosion.

* 3.6.3 Flammability. The cable assembly shall meet the requirements as specified in the specification sheet.

* 3.6.4 Deleted.

3.6.5 Vibration. All cable assemblies which contain connectors, which in use, are hard-mounted to the aircraft, shall meet the requirements of 3.5.3 while tested in accordance with 4.7. Only the connector and a section of its associated cable need be tested.

3.6.6 Environmental service conditions. Each cable assembly shall meet the specified performance requirements of the applicable specification sheet and 3.5.3, within one hour of being subjected to the following environmental tests of 4.7:

a. Shock test

b. Humidity

c. Thermal shock

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The cable jacketing shall show no signs of cracking after completion of the Thermal Shock Test.

3.7 Workmanship. The cables shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, functioning, and appearance. Cleaning shall be in accordance with MIL-STD-454, Requirement 9. There shall be no evidence of loose electrical connectors; poor or improper molding or fabrication; damaged or improperly assembled contacts; peeling; flaking or chipping of plating or finish; mechanical damage due to testing environment; nicks or burrs of metal parts of surfaces; improper or incorrect marking; or improper tinning of solder cups, terminals, pins, or contacts.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Qualification inspection (see 4.4).

Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph(s) of this specification.

4.3.1 Test equipment. All instruments used in performing inspections required by this specification shall be maintained and calibrated in accordance with MIL-STD-45662 and shall, as a minimum, meet the following requirements. The intervals between calibration shall be no greater than one

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year. Any test equipment not meeting the following criteria must be approved by the Qualifying Activity (Naval Avionics Center, Code 444) prior to use in performing the inspections required herein.

* 4.3.1.1 Multimeter.

a. Digital readout (4-1/2 digits)

b. Direct current (dc) and true RMS alternating current (ac)

c. Ranges: 10 microvolts to 15 volts RMS ac
1 to 30 volts dc 1/

d. Accuracy: ±.5%

e. Input impedance: 1 megohm minimum

f. Inputs isolated from ground up to 30 volts

1/ NOTE: A calibrated lock-in amplifier with fixed incremental gain control may be used as a true RMS ac voltmeter provided all other requirements are met. A range of 10 millivolts to 15 volts is acceptable providing the attenuator maintains calibration under varying loads over the frequency range.

4.3.1.2 Signal or function generator.

a. Sine wave output

b. Frequency range: 100 to 6000 Hz minimum

c. Frequency accuracy and flatness at full output: ± 2 percent

d. Harmonic distortion: Less than -65 dB (relative to fundamental) at full output for each frequency

e. dc offset: 1 millivolt maximum

f. Voltage range: 1 millivolt to 15 volts 2/

g. Output isolated from ground up to 30 volts

2/ NOTE: A suitable amplifier may be used providing the function generator plus amplifier combination meets the above requirements.

4.3.1.3 Spectrum analyzer (optional).

a. Frequency range: 100 to 6000 Hz

b. Center frequency accuracy: ±.003%

c. Bandwidth: 1 Hz minimum to 500 Hz maximum

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d. Measurement ranges: 1 microvolt to 15 volts linear
-120 dBV to +30 dBV log

e. Dynamic range: 70 dB

f. Voltage accuracy: ±.5 dB

g. Input impedance: 1 megohm minimum

h. Inputs isolated from ground up to 30 volts

4.3.1.4 Distortion analyzer.

a. Frequency range: 100 to 6000 Hz

b. Frequency accuracy: ±2%

c. Measurement range: 30 microvolts to 15 volts

d. Dynamic range: 70 dB

e. Voltage accuracy: ±2%

f. Input impedance: .1 megohm minimum

g. Fundamental rejection: 70 dB

h. Inputs isolated from ground up to 30 volts

4.3.1.5 Precision decade resistance box.

a. Frequency range: dc to 6000 Hz

b. Resistance range: 2 to 15 ohms

c. Resistance increments: .1 ohms

d. Accuracy: .5%

e. Resistance zero setting: 12 milliohms or less

f. Inductance: Less than .03 microhenries

g. Capacitance: Less than 8 picofarads

h. Low temperature and power coefficients

4.3.1.6 Lock-in amplifier (optional).

a. Frequency range: 100 to 6000 Hz

b. Sensitivity: 1 microvolt to .5 volts

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c. Output time constants: 1 millisecond to 10 seconds

d. Harmonic rejection: 55 dB

e. Normal output stability: 50 ppm

f. Gain accuracy: ±2%

g. Digital readout (3-1/2 digits)

h. Phasing: Automatic

i. Input impedance: 1 megohm or greater

j. Gain: Incremental or lockable after calibration

4.3.1.7 Oscilloscope.

a. Channel: Single

b. Bandwidth: 5 megahertz

c. Time base: 20 microsec/div to 1 millisec/div

d. Time base accuracy: ±3%

e. Coupling: ac and dc

f. Source: Internal

g. Sweep mode: Auto trigger

h. Voltage base: 5 millivolts/div to 10 volts/div

i. Voltage base accuracy: ±3%

j. Input impedance: 1 megohm minimum

4.3.1.8 Ac Power Supply (optional).

a. Sine wave output

b. Frequency range: 100 to 6000 Hz minimum

c. Frequency accuracy and flatness at full output: ±2%

d. Harmonic distortion: Less than -65 dB (relative to fundamental) at full output for each frequency

e. dc offset: 1 millivolt maximum

f. Voltage range: 1 millivolt to 15 volts

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g. Output power: 220 VA with continuous volt-ampere output even under zero or unity power factor loads

h. Output isolated from ground up to 30 volts

* 4.3.1.9 AC ratio transformer standard (see 6.4).

a. Ratio: 1.111111 maximum, 0.0111111 minimum.

b. Frequency range: 50 Hz to 10 KHz.

c. Terminal linearity: 0.0001% (1 ppm).

d. Resolution: 0.00001% steps.

e. Maximum effective series impedance: R series 3.5 L series 100 microhenries.

f. Output isolated from the input.

* 4.4 Qualification inspection. Qualification inspection shall be performed in accordance with the provisions of SD-6. The qualification inspection shall consist of all Group I tests specified in Table I of the specification sheet for each cable assembly. The qualification inspection sample shall consist of fifteen units of each cable assembly for which qualification is desired. At least two of the fifteen units shall be considered control samples, and only Group IA tests shall be performed on these units until the event of a failure. Group IA inspections shall be performed on each of the fifteen sample items. At least two samples shall be submitted to each of the Group I (B, C and D) inspections. In the event of a single failure, one of the control samples shall be submitted to the failed test sequence. If a second failure occurs, qualification shall be denied.

4.4.1 Deleted.

* 4.4.2 Items for qualification must be separately packaged and forwarded to the activity responsible for qualification as designated in the letter of authorization. The sample shall in all respects be representative of the manufacturer's production item. Each of the items shall be subjected to the applicable inspections specified in Group I of Table I of the specification sheet. Samples shall be marked in accordance with 3.4.19 except the items shall also be marked (Qualification Samples) and shall include the applicable specification number, and shall reference the letter authorizing the tests.

4.4.3 Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Acceptance tests shall consist of Group II and Group III tests in Table I of the specification sheet for the specified cable assembly.

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* 4.4.4 Retention of qualification. Contractors wishing to remain on the Qualified Products List shall submit to the preparing activity at intervals no greater than three years, DD Form 1718, Certification of Qualified Products. At the discretion of the qualifying activity, fifteen (15) Qualification samples shall be submitted to the QPL agent for verification of design requirements. The QPL agent should perform the minimum retention tests noted in the specification sheet to allow retention of qualification, and any other tests deemed necessary to insure that the basic design requirements for the cable assembly are still being met. The contractor may submit production units as Qualification retention samples.

4.5 Quality conformance inspection. Quality conformance inspection for this specification shall consist of the tests of Group II and Group III of the specification sheet unless otherwise specified. Group II quality conformance tests shall be performed on each individual unit prior to acceptance. Units failing the quality conformance test may be discarded or reworked to correct defects and resubmitted for acceptance.

4.5.1 Production lot sampling. All cable assemblies submitted for inspection as a production lot shall be manufactured using identical components, raw materials and processes. All cordage and molding material shall be from the same incoming lots. All other components of identical type designation shall be procured from the same manufacturer for all cable assemblies in the production lot (the manufacturer shall not change cordage manufacturers, or connector manufacturers, etc., during a production lot). All Group III tests specified in Table I of the specification sheet shall be performed on sample units pulled from the production lot which has already passed the Group II inspection. The sample units shall be chosen at random under the supervision of the cognizant government inspector. Sample sizes for Group III sample testing as specified in Table I of the specification sheet shall be in accordance with Table I of the general specification. Contracts for less than 10 cables shall be subjected to 100% inspection of Group A tests.

TABLE I. Sample plan for Group III sample testing.

Number of Cable Assemblies in Contract	Total Number of Cable Assemblies to be Tested
10 thru 280	3
281 thru 500	6
501 thru 1000	9
1001 thru 2000	12
2001 thru 5000	15
5001 or more	18

4.5.1.1 Disposition of samples. Samples subjected to Group B tests may be shipped under the contract. Samples subjected to Group C tests shall not be shipped under the contract.

4.5.2 Contractors not having laboratory testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the Naval Avionics Center (NAC), Indianapolis, Indiana.

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4.6 Methods of examination and test.

4.6.1 Test conditions. Unless otherwise specified herein or in the equipment specification, measurements and tests shall be made at ambient conditions.

4.6.2 Visual and mechanical examinations. The cable assemblies shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements.

4.6.2.1 Soldering. All soldering shall meet the general purpose soldering requirements of DOD-STD-2000.

* 4.6.3 Deleted (see MIL-C-22442/1).

* 4.6.3.1 Deleted (see MIL-C-22442/1).

* 4.6.3.2 Deleted (see MIL-C-22442/1).

* 4.6.3.3 Deleted (see MIL-C-22442/1).

* 4.6.3.4 Deleted (see MIL-C-22442/1).

* 4.6.4 Deleted (see MIL-C-22442/1).

* 4.6.4.1 Deleted (see MIL-C-22442/1).

* 4.6.4.2 Deleted (see MIL-C-22442/1).

4.6.4.3 Deleted (see MIL-C-22442/1).

4.6.4.4 Deleted (see MIL-C-22442/1).

4.6.4.5 Deleted (see MIL-C-22442/1).

4.6.4.6 Deleted (see MIL-C-22442/1).

4.6.5 Continuity. dc resistance checks shall be made in accordance with the specification sheet. All conductors terminating in active components shall be tested before the active component is included in the circuit.

4.6.6 Deleted.

* 4.6.7 Dielectric withstanding voltage. Dielectric strength shall be performed in accordance with Method 301 of MIL-STD-202. The potential and time durations shall be as specified in the applicable cordage specification. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. All conductors terminating in active components shall be tested before the active component is included in the circuit.

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4.6.8 Gauging. Contact locations and interfacial dimensions of the connector as specified in the applicable connector specification or drawing, shall be measured after assembling connector to cable. Certified test data from the connector manufacturer may be used in lieu of final assembly gauging. It is mandatory that the cable manufacturer perform gauging on the molded rubber connector (M22442/2-1 and M22442/2-2).

* 4.6.9 Pull and twist test. The cable assembly pull test shall consist of the twist test (4.6.9.1) followed by the tensile test (4.6.9.2).

4.6.9.1 Twist test. The twist shall be applicable to all ends of the cable assembly that are terminated with a connector only. Each connector shall be secured in a holding device in such a way that it cannot rotate. The twist shall be applied about the axis of the cable to cause a point on the cable three inches from the connector to twist 180 degrees from the starting position and be held there for 30 seconds. The cable shall then be twisted in the opposite direction such that the point on the cable has twisted 180 degrees past the original starting position, and be held there for 30 seconds.

* 4.6.9.2 Pull test. The pull test shall be applied to all terminals, connectors, switch assemblies and taper pins. The component of the cable assembly to be tested shall be secured in a stationary holding device and the cable gripped six to twelve inches from the component of the cable assembly under test. The grips shall not slip or damage the cable during the test. The static force shall be applied for 30 seconds with the force values specified below.

4.6.9.2.1 Terminals. Terminals soldered to circular conductors shall be subjected to 5 pounds minimum pull. Terminals crimped to circular conductors shall be subjected to 3 pounds minimum pull. Contacts as part of the connector shall not be considered a terminal.

4.6.9.2.2 Taper-pins. Taper-pins inserted into a connector shall be subjected to a pull of 3 pounds minimum at the taper-pin connector connection.

4.6.9.2.3 Connectors. A pull force of 20 pounds shall be applied; 10 pounds more shall be added if the connector is potted or molded with bonded bend reliefs or bonded strain-reliefs.

4.6.9.2.4 Switch assembly. The switch assembly shall be securely mounted and a 20 ± 2 lb. force shall be applied to the connecting cables. The cable assembly shall meet the requirements of 3.5.7.

4.6.10 Deleted.

4.6.11 Deleted.

* 4.6.12 Bend relief flexibility. Each bend-relief in the cable assembly shall be manually subjected to 30 continuous flexure cycles, at room temperature, through a 180 degree arc, at approximately 30 cycles per minute. Each bend relief shall be held in a mechanical device so that the portion of the cable entering the bend relief flexes when the cable is flexed. The bend-relief is free to flex with the cable in one direction through 90 degrees

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from the axially in-line starting position, back to the starting position, then through 90 degrees in the opposite direction, and return to the starting position. The flexing procedure shall be repeated for 30 cycles in a plane 90 degrees axially to the plane of initial flexing.

4.6.13 Low temperature flexibility. The cable assembly or portion thereof containing the molded section (see 6.5) or heat-shrunken transition (see 6.5) shall be aged in a circulating air oven for a minimum of 48 hours at 160 ± 2°F. The cable assembly shall then be conditioned for a minimum of 8 hours at the temperature specified in 3.5.11. This temperature shall be continuously recorded during conditioning and testing. Each crotch or molded section (see 6.5) shall be held in a mechanical device so that the portion of the cable entering the molded section flexes when the cable is flexed. The cable shall be flexed mechanically. Any part of a mechanical device which will be in contact with the test specimen shall be cold conditioned. At the end of the conditioning period and while at the conditioning temperature, the cable shall be flexed rapidly for 30 cycles through a 180-degree arc (± 90 degrees from 0 degrees). The flexing procedure shall be repeated in a plane 90 degrees to the plane of initial flexing. At the conclusion of the test, the specimen shall be examined visually at room temperature under 3X magnification.

4.6.14 Deleted.

4.6.15 Durability. All connectors of the cable assembly shall be mated and unmated 100 times unless otherwise specified. MIL-C-26482 type connectors shall be mated manually or automatically at a rate not to exceed 12 cycles per minute, and shall be torqued to the value specified in the applicable MIL-C-26482 specification sheet. Quick disconnect connectors shall be unmated by breaking the connectors in a direction perpendicular to the axial plane and major axis. All other connector types shall be mated and unmated in the axial direction, at a rate not to exceed 12 cycles per minute.

4.6.16 Ozone resistance. The cable assembly shall be tested for ozone resistance in accordance with ASTM D1149 with the following test conditions: Test time shall be 168 ± 1 hour; ozone concentration shall be 50 parts per 100,000,000; and the chamber temperature shall be 100 ± 5°F. Any bend relief or a cordage intersecting a molded junction shall be bent at an angle of 90 degrees over the mandrel while in the chamber. The mandrel radius shall be based on the cross sectional area of the bend relief or a cordage intersecting a molded junction.

Specimen Cross Section	Mandrel Radius
Up to 0.500	4 x Cross Section
0.501 to 0.750	5 x Cross Section
0.751 to 1.250	6 x Cross Section

4.6.17 Jacket elongation. The elongation test shall be in accordance with Method 3031 of the FED-STD-228.

* 4.6.18 Durometer hardness. Durometer hardness shall be measured by use of a durometer hardness gauge acceptable to the government inspector. The tolerance shall be ± 10 shore A units.

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4.6.19 Material reversion. The molding, potting, insulating, and jacketing material shall be tested using specimens approximately 4 inches square by 1/8 inch thick, (6.5 inches for jacketing material) for Durometer Hardness Shore "A". Then, they shall be placed in a chamber maintained at 100 + 1°C and 95 + 2 percent relative humidity without condensation for 28 days. The cable assemblies shall be arranged so they do not come in contact with each other nor with the chamber walls or floor. After conditioning, the cable assemblies shall be tested in accordance with 3.5.17.

4.6.20 Deleted.

4.6.21 Deleted.

4.6.22 Deleted.

* 4.7 Environmental service condition tests. The cable assemblies shall meet the performance requirements of 3.6 when tested in accordance with the following procedures.

a.	Temperature-altitude	MIL-T-5422, Class I 1/
b.	Salt fog	MIL-T-5422
c.	Flammability	UL-1581
d.	Vibration	MIL-T-5422, Procedure II, Parts I and II (Figure 5, Curve IV and Figure 7, Curve VIII)
e.	Shock	MIL-T-5422, Part I (Basic Design test only)
f.	Humidity	MIL-T-5422
g.	Thermal shock	MIL-T-5422

1/ Steps 1b, 3, 4, 5, 7 and 12 only unless otherwise specified in the specification sheet.

5. PACKAGING

5.1 Packaging. Cable assemblies shall be packed in accordance with MIL-C-55442. The level of protection shall be as specified in the contract.

5.2 Marking. Unless otherwise specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cable assemblies covered by this specification are intended for use in any application where their performance characteristics are required. The cables are suitable for installation on aircraft systems within the limitations of applicable performance requirements.

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6.1.2 Materials compatibility. The jacketing systems of the cable assemblies covered by this specification may be degraded by certain fluids or compounds. If such degradation occurs, the fluids or compounds and the conditions necessary for failure shall be added to the specification sheet as a precautionary note.

6.2 Ordering Data. Procurement documents should specify the following:

6.2.1 Title, number, and date of this specification.

6.2.2 Applicable specification sheet number, title, and date.

6.2.3 Applicable specification sheet part number.

6.2.4 Accessories, when they are desired to be furnished with the cable assemblies.

6.2.5 Number of cable assemblies required.

6.2.6 Level of packaging and packing required.

6.2.7 Exceptions, if any to the optional provisions of this specification including:

6.2.7.1 Responsibility for inspection, if other than specified (Section 4).

6.2.7.2 Special preparation for delivery requirements, if applicable (Section 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.3.1 Conformity to Qualified Sample. It is understood that cable assemblies supplied under contract shall be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

6.3.2 Application for Qualification. The activity responsible for the QPL for MIL-C-22442 is the Naval Air Systems Command. The Naval Avionics Center, Indianapolis, Indiana, has been designated by the Naval Air Systems Command as agent for the establishment and maintenance of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

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Commanding Officer
Naval Avionics Center
Code 444
6000 East 21st Street
Indianapolis, IN 46219-2189

6.4 Rationale. The following historical documentation provides rationale for various requirements in this specification.

6.4.1 Test equipment. The test equipment described in 4.6 was selected to minimize testing problems encountered in measuring low level AC true RMS voltages and impedances. The low level AC microvoltages are several orders of magnitude lower in value than the thermally induced voltages. The low level impedances are equivalent to short circuits for some instruments and thereby create test induced measurement errors. Although automated test equipment is permitted and encouraged, be aware of test induced problems created by various analogue to digital conversion techniques and aliasing problems. These test induced problems include the generation of nonexistent failures as well as the concealment of bonafide failures.

6.4.2 Figure 8. For MIL-C-22442/1, the amplifier input voltage range of 50 to 500 microvolts RMS creates NBS traceability problems for calibrated AC true RMS voltmeters. Many multimeters and true RMS voltmeters have unwarranted claims as to an AC true microvoltage measuring capability. On the lowest range scale, the instruments do not maintain the measurement nor calibration accuracy that is true at higher range scales. The instrument specification sheet itself is often unclear as to the actual accuracy when measuring a microvolt signal. The attenuation circuit of Figure 8 can resolve this problem without the use of an expensive AC true RMS microvoltmeter provided the following conditions are met:

a. The attenuation value is calibrated with NBS traceability at measurable voltage levels.

b. The impedances do match between the function generator and the amplifier under test.

c. The attenuation value is 80 dB and not 60 dB.

NOTE: EXPERIMENTAL EVIDENCE SHOWS THAT A 60 DB ATTENUATOR DOES NOT PROVIDE A CONSTANT VOLTAGE INDEPENDENT OF FREQUENCY OR ANTICIPATED LOAD IMPEDANCES.

An AC ratio transformer standard (see 4.6.8), whose ratio is calibrated at a higher voltage level, can serve as an adequate transfer standard for verifying an instrument with AC true RMS microvoltage measuring capability and for calibrating the attenuator described in Figure 8.

6.5 Definitions.

Bend relief. Any method used to alleviate stress on a section of cable due to bending of the cable. Examples: Adding rubber at the cable/connector interface to reduce cable stress caused by movement of the connector.

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Bonded bend relief. A bend relief which is molded to the connector.

Bonded strain relief. A strain relief which is molded to the cable assembly.

Crotch. The point at which one cable separates into two or more branches.

Heat shrunken transition. Placing a piece of heat shrinkable tubing around a portion of cable and shrinking in place, creating a heat shrunken transition. This is usually placed around an interface. (Example: Where the connector meets the cable.)

Molded junction. Encapsulating in rubber a junction of one or more cables, or wires, creates a molded junction.

Molded section. A molded section is created by encapsulating a portion of the cable jacket in rubber. This differs from the molded junction in that it is not limited to cable junctions.

Potted connector. A connector which is sealed by using a potting or encapsulating compound on the backside.

6.6 Subject term (key word listing).

Audio cable
Cable assemblies
Cord sets

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DLA - ES

Preparing Activity:
Navy - AS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22442D(AS)	CABLE ASSEMBLIES, RADIO FREQUENCY (AIRCRAFT AUDIO), GEN. SPEC. FO

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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